Modern Delivery

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Database Management Course

Summary:

1. Introduction
2. Database Structure
3. Database Connectivity
4. Error handling and Optimization
5. Application Demo
6. Conclusion

1. Introduction

Problem:

-Logistics systems involve ships, ports, and routes

-Raw data is difficult to interpret without visualization

-Need for a system combining database + map interface

Objectives:

-Visualize ships and seaports on a map

-Connect to a PostGIS database

-Provide search and navigation tools

Technologies/librairies:

-Python + PySide6 (UI)

-PostgreSQL + PostGIS

-OpenStreetMap + Nominatim API

2. Database Structure

ER-diagram

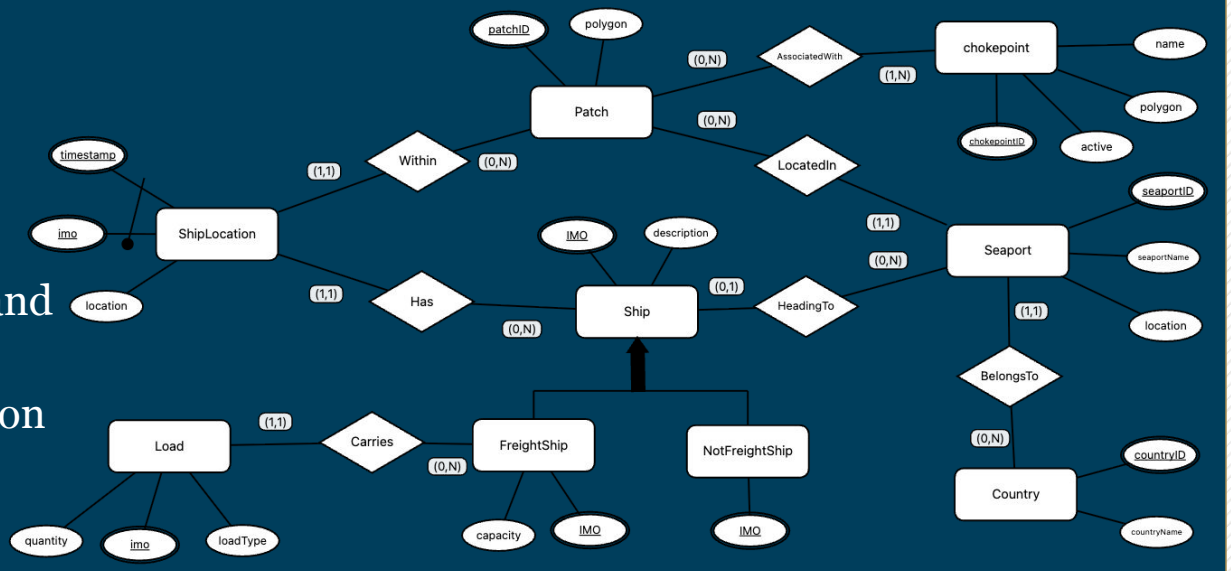
Main tables:

Seaports: stores ports linked to countries and their locations

Ships: stores ship imos and descriptions

Chokepoints: store maritime bottlenecks identified by name and activity status

Patches: store geographic polygon areas linked to chokepoints



3. Database Connectivity

Program layers:

-Data Layer: db_connector (queries, connection)
-Logic Layer: ItemManager + models (Ship, Seaport, Node)
-UI Layer: Map view and widgets

Data Flow:

Database → Connector → ItemManager → Map

Spatial Processing:

-Use of PostGIS functions (e.g., ST_VoronoiPolygons)
-Allows spatial partitioning of areas

4. Error Handling and Optimization

Database Error Handling:

-Managing data quality via enforcing constraints

-Controlled extension set up

-Implementation of spatial validation logic

Performance Optimization:

-Use of database indexes

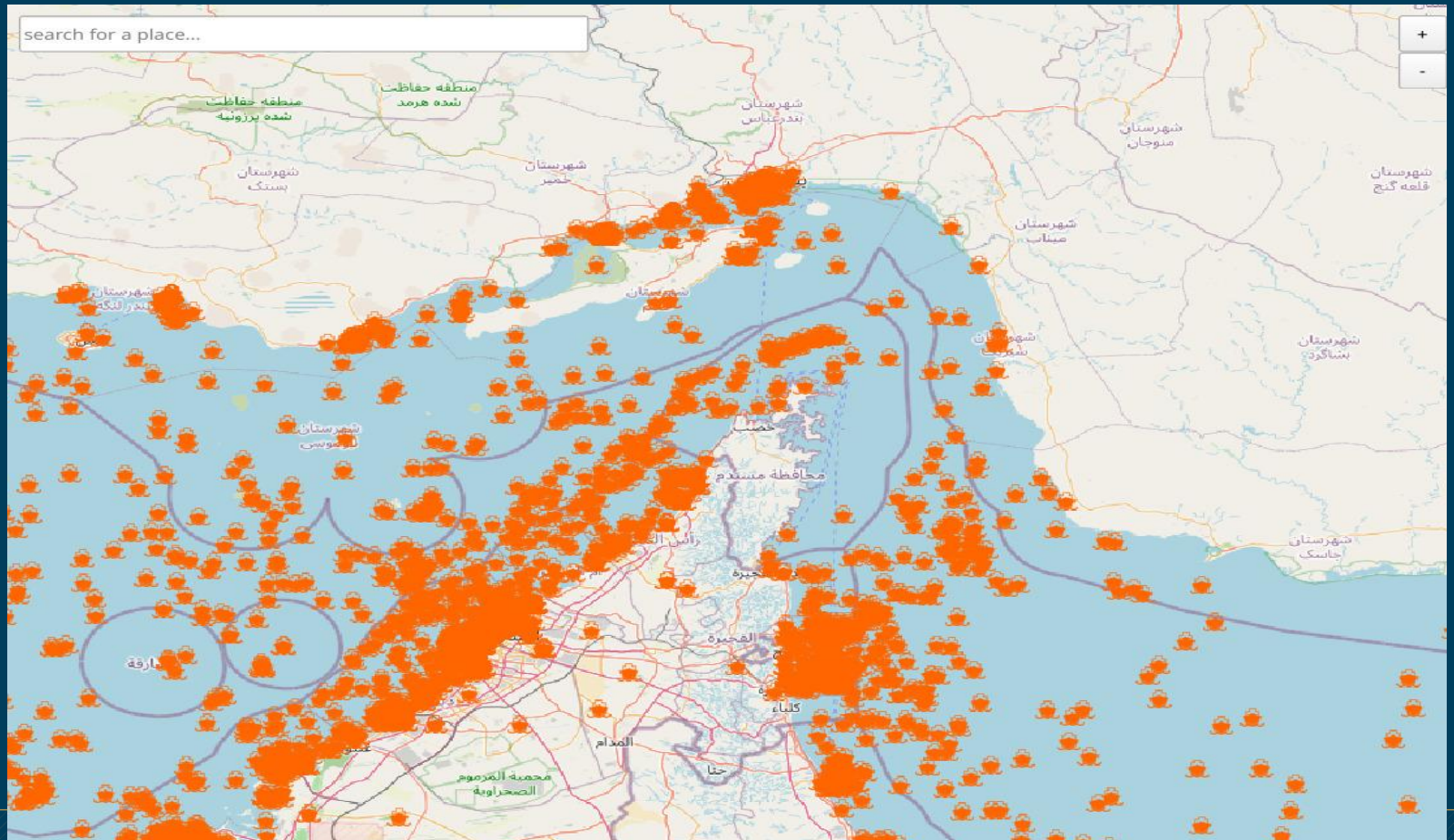
-Spatial indexing for ship and node locations

-Separation of current ship location (state_ships) and historical ship locations

Application Stability:

-Handling potential concurrency issues by introducing a trigger and a function

5. Application demo



6. Conclusion

Strengths:

-Clear architecture and modular design
-Strong integration of PostGIS
-Functional map visualization system

Future Improvements:

-Animate ships movement
- Improve interaction (selection, routes)
-Real-time updates
-Implement a more sophisticated rerouting logic

Merci

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